Summative Project
Digital Electronics 2B EEDE226
semester 2

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Title: Gas Detection and Alert system using STM32 NUCLEO-F401R3

Introduction

This project involves designing and implementing a gas detection and alert system using the STM32 NUCLEO-F401RE microcontroller. The system detects harmful gases in environment using a MQ15 gas sensor displays the air quality status on the 16x2 $\mathrm{I^2C}$ LCD and activates a buzzer alarm when dangerous gas levels are detected.

Components used

Component	Function
STM32 NUCLEO-F401RE	Used for processing and control
MQ135 Gas sensor	Detects gases such as $C0_2, NH_2$Benzene and Alcohol
ADS1115 ADC module	16 bits analog to digital converter for precise analog signal reading
16x2 LCD I^2C	Display system status (safe or Gas alert)
HC-05 Bluetooth Module	Sends data wirelessly
NPN transistor (2N2222)	Drives the buzzer circuit
Buzzer	Emits sound when gas concentration is high
Resistor	Used for voltage divider and signal conditioning

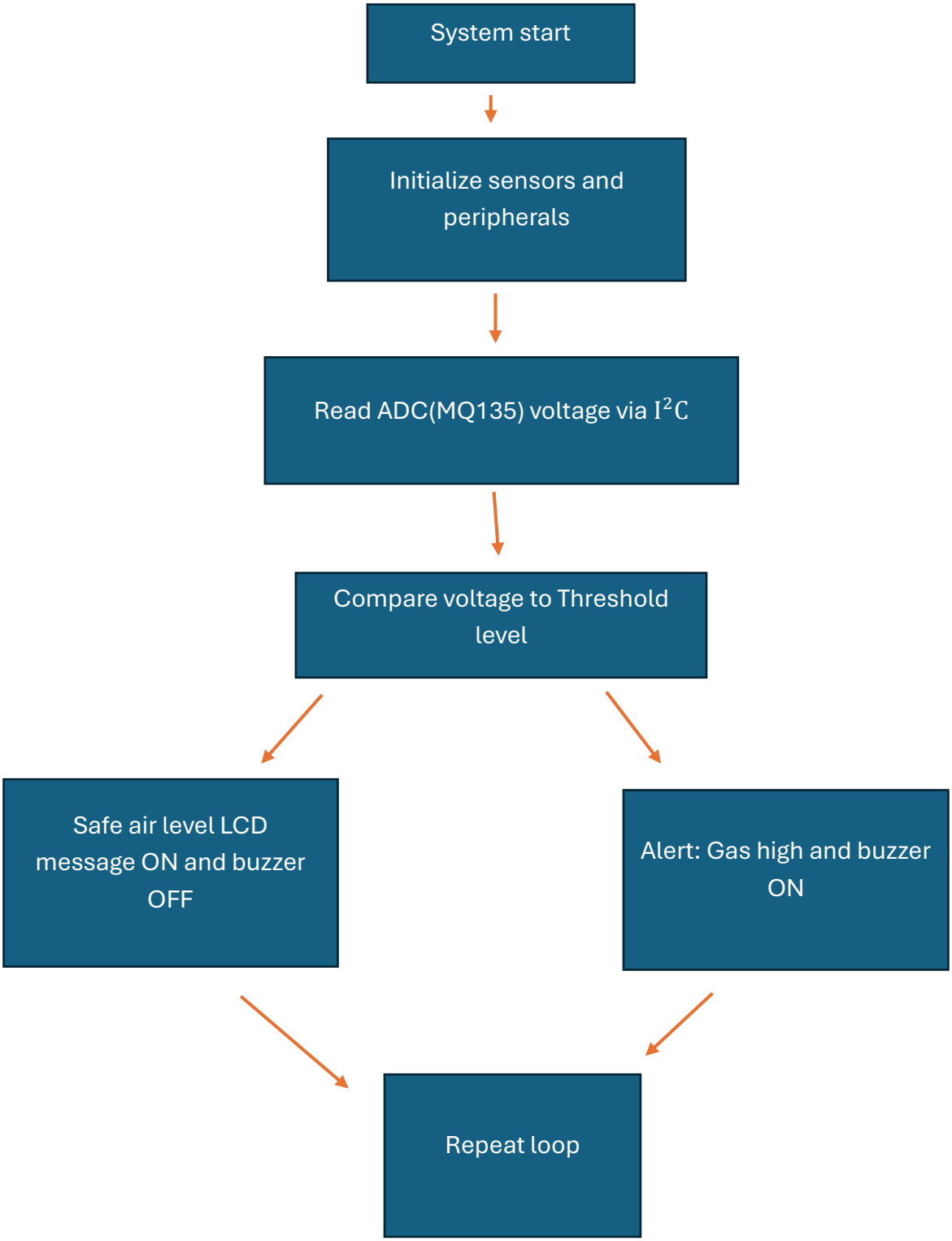
System design and operation

The system is designed around the STM32 NUCLEO-F401RE microcontroller, which serves as the main controller for sensing, processing, and displaying gas concentration data. The MQ135 gas sensor continuously monitors the air quality and outputs an analog voltage that varies with the detected gas concentration. This analog signal is fed into the ADS1115 analog-to-digital converter (ADC), which converts it into a digital value readable by the STM32 via the $\mathrm{I^{2C}}$ communication interface.

After receiving the digital value, the STM32 processes the data and compares it to a predefined threshold voltage. If the voltage corresponds to a safe gas level, the system displays **"**SAFE AIR LEVEL**"** on the LCD and ensures the buzzer remains off. If the voltage exceeds the threshold, indicating a high gas concentration, the system displays **"**ALERT: GAS HIGH**"** and activates the buzzer alarm. A transistor driver circuit is used to control the buzzer, ensuring it receives sufficient current without damaging the STM32 output pin.

The LCD and ADS1115 modules are both connected using I^2C communication, with separate address lines to avoid conflicts. The system operates on a 3.3V supply from the STM32 board, ensuring safe voltage compatibility for all components

System Flow diagram



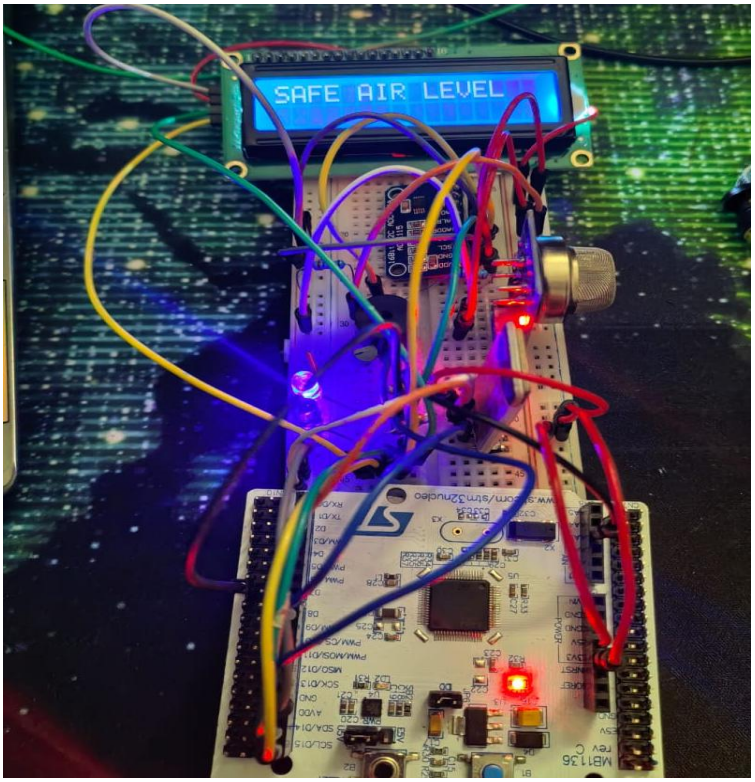
STM32 system Hardware connection and configuration

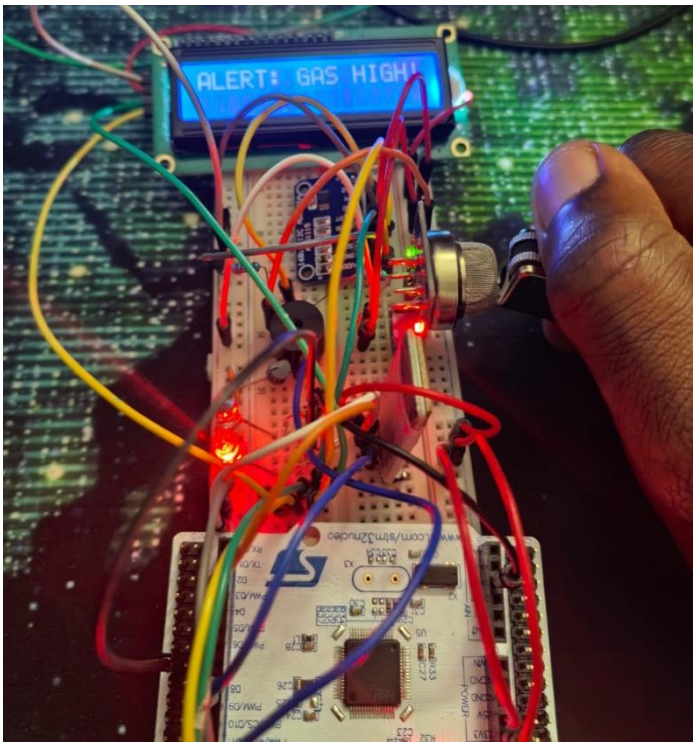
Component	Pin	Connected to STM32 pins	function	Purpose
MQ135 gas sensor	Vcc	3.3V	Power supply	Power the gas sensor
	GND	GND	Ground	Common ground connection
	AOUT	ADC Input (ADS1115 A0	Analog Output	Sends analog voltage representing gas concentration
	DOUT	Not Connected	Digital Output	Unused digital comparator output
	Potentiometer	On-board	Adjustable Threshold	Adjusts gas detection sensitivity
ADS1115 ADC Module	VCC	3.3V	power	Power ADC
	GND	DNG	Ground	Common reference
	SDA	PB9	I2C1_SDA	Data line to STM32
	SCL	PB8	I2C1_SCL	Clock line to STM32
	A0		Analog input	Reads gas sensor voltage
	A1, A2, A3,		Not connected	Unused channels
	ALERT		Not connected	Optional interrupt pin
LCD I2C 16X2	Vcc	5V	Power supply	LCD power
	GND	GND	Ground	Common reference
	SDA	PB9	I2C1_SDA	Data communication
	SCL	PB8	I2C1_SCL	Clock communication
Buzzer via Transistor	+V	5V	Power	Power supply for buzzer
	Base	GPIO pin 5 via resistor	Control signal	Turns buzzer ON/OFF
	Collector	Buzzer negative terminal	Load connection	Connects buzzer circuit
	Emitter	GND	ground	~
LED indicator	Red LED (anode)	PB1 (via 220 Ω resistor)	GPIO output	ON = High gas concentration
	Red Led (cathode)	GND	Ground	~
	Blue LED (anode)	PB0 (via 220 Ω resistor)	GPIO output	ON = safe air level

	Blue LED (cathode)	GND	Ground	~
Bluetooth module (HC-05)	Vcc	3.3V	Power	Power
	GND	GND	Ground	~
	TX	PA10	UART RX	Sends data to ST32
	RX	PA9	UART TX	Receives data from STM32
	EN, STATE	Not connected	~	~

Results and testing

During testing, the system successfully detected gas presence using the MQ135 sensor and responded according to concentration levels. When no gas was present, the LCD displayed **"SAFE AIR LEVEL"** and the buzzer remained off. When gas concentration increased beyond the set threshold, the display changed to **"ALERT: GAS HIGH"** and the buzzer activated, signalling danger.





Conclusion

The Gas detection and alert system was successfully designed and implemented. It effectively detects the presence of harmful gases and provides both visual (LCD) and sound (buzzer) alerts. This project demonstrates the integration of analog sensor, digital

conversion and i2c communication using STM32 NUCLEO-F401RE.future improvements could include mobile alerts.

References

1. STM32F4 Series Reference Manual – STMicroelectronics
2. MQ135 Gas Sensor Datasheet – Rajguru Electronics
3. ADS1115 ADC Datasheet – Texas Instruments
4. STM32CubeIDE Documentation – STMicroelectronics
5. HC-05 Bluetooth Module datasheet- ITead Studio